LECTURE 24

22 November 2023 08:37

Lagrange's Multiplier Method: Constrained optimization: Two variables & 1 constraint.

problem: Optimize $f(x,y)$ subject to $g(x,y) = 0$

Steps: 1) Lagrangian $f^n$: $L(x,y,\lambda) = f(x,y) + \lambda g(x,y)$

Lagrange Multiplier

2) $\frac{\partial L}{\partial x} = 0$, $\frac{\partial L}{\partial y} = 0$, $\frac{\partial L}{\partial \lambda} = 0$

3) Use obtained $eq^n$ to find $\lambda$.

4) Use that $\lambda$ to find stationary point $(x,y)$.

5) Put these points in $f(x,y)$.

Q. 1) Max $f(x,y) = x+y$ subject to $x^2+y^2 = 1$

$x^2+y^2-1 = 0$

$g(x,y) = 0$

$L(x,y,\lambda) = x+y+\lambda(x^2+y^2-1)$

$\frac{\partial L}{\partial x} = 1+0+\lambda \cdot 2x = 1+2\lambda x = 0$

$\frac{\partial L}{\partial y} = 1+\lambda 2y = 1+2\lambda y = 0$

$\frac{\partial L}{\partial \lambda} = x^2+y^2-1 = 0$

$1+2\lambda x = 0$ (1)

$1+2\lambda y = 0$ (2)

$x^2+y^2 = 1$ (3)

(1) $\Rightarrow x = \frac{-1}{2\lambda}$, (2) $\Rightarrow y = \frac{-1}{2\lambda}$ Putting these in (3)

$\frac{1}{4\lambda^2} + \frac{1}{4\lambda^2} = 1 \Rightarrow \frac{2}{4\lambda^2} = 1$

$\Rightarrow \lambda = \pm \frac{1}{\sqrt{2}}$

$x = y = -\frac{1}{2\lambda}$

a) $(x,y) = \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

b) $(x,y) = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

$\lambda = -\frac{1}{\sqrt{2}}$

For $\lambda = +\frac{1}{\sqrt{2}}$

$-\sqrt{2} - \sqrt{2} = 2 \cdot \frac{-\sqrt{2}}{2} = -\sqrt{2}$

For $\lambda = \frac{\sqrt{2}}{2}$

$$f\left(\frac{-\sqrt{2}}{2}, \frac{-\sqrt{2}}{2}\right) = \frac{-\sqrt{2}}{2} - \frac{\sqrt{2}}{2} = 2 \cdot \frac{-\sqrt{2}}{2} = -\sqrt{2}$$

$$f\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right) = \sqrt{2}$$

Q 2b) Max. $f(x,y) = x^2 y$ s.t. $x^2 + y^2 - 3 = 0$

$$L(x,y,\lambda) = x^2 y + \lambda(x^2 + y^2 - 3)$$

$$\frac{\partial L}{\partial x} = 2xy + \lambda \cdot 2x = 2x(\lambda + y)$$

$$\frac{\partial L}{\partial y} = x^2 + \lambda \cdot 2y = x^2 = -2\lambda y$$

$$\frac{\partial L}{\partial \lambda} = x^2 + y^2 - 3$$

$x(\lambda + y) = 0$ 1)

$x^2 = -2\lambda y$ 2)

$x^2 + y^2 = 3$ 3)

1 $\Rightarrow$ Either $x = 0$ or $y = -\lambda$

case 1) when $x = 0$ 3 $\Rightarrow y = \pm\sqrt{3}$ $\lambda = 0$ a) $(0, \sqrt{3})$ b) $(0, -\sqrt{3})$

case 2) when $y = -\lambda$ 2 & 3 $2\lambda^2 + \lambda^2 = 3 \Rightarrow 3\lambda^2 = 3$

$\boxed{\lambda = \pm 1}$

$y = 1$ or $-1$

3 $\Rightarrow$ $x = \sqrt{2}$ or $-\sqrt{2}$

c) $(\sqrt{2}, 1)$ d) $(\sqrt{2}, -1)$ e) $(-\sqrt{2}, 1)$ f) $(-\sqrt{2}, -1)$

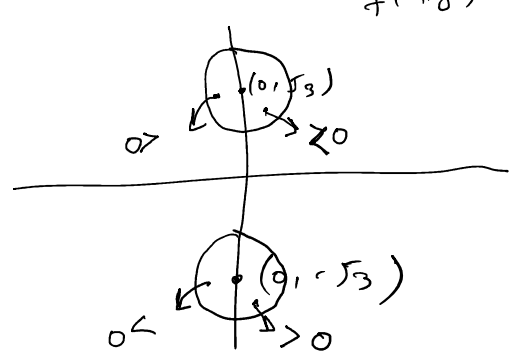
case $f(\pm\sqrt{2}, 1) = 2$ max. $f(\pm\sqrt{2}, -1) = -2$ mini.

(a&b $f(0, \pm\sqrt{3}) = 0$

at $(0, \sqrt{3})$ $\rightarrow$ local mini.

$(0, -\sqrt{3})$ local max.

$f(x,y) = x^2 y$



2.3. minimize $f(x,y) = x^2 + y^2$ s.t. $x + y = 1$